

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	PRITHIYANKA SHREE M	Reg. No.:	192521144
Section / Batch:	2025-IT	Date:	29/07/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Linear Search – Target in Second Half

Given the unsorted dataset [61, 24, 13, 47, 32, 89, 53], apply Linear Search to find the element 89. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required to locate the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target appears near the end of the dataset.

Code of Conduct

I PRITHIYANKA SHREE M (192521144) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: PRITHIYANKA SHREE M

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Unsorted array: $[61, 24, 13, 47, 32, 89, 53]$

The task is to find the element 89 using the Linear Search algorithm.

Algorithm:

- Start from first element.
- Compare each element with the target (89).
- If the element matches the target, stop the search.
- Otherwise, move to the next element.
- Continue until the target is found or the array ends.

step	Element compared	comparison	Result.
1.	61	$61 = 89?$	No
2.	24	$24 = 89?$	No
3.	13	$13 = 89?$	No
4.	47	$47 = 89?$	No
5.	32	$32 = 89?$	No
6.	89	$89 = 89?$	Yes(found)

Number of Comparisons:

61 $\rightarrow$ 1

24 $\rightarrow$ 2

13 $\rightarrow$ 3

47 $\rightarrow$ 4

32 $\rightarrow$ 5

89 $\rightarrow$ 6

Total comparisons = 6

Verification:

Array: [61, 24, 13, 47, 32, 89, 53]

Target element: 89.

Position:

Index = 5 (0-based indexing)

Hence, the linear search correctly locates the target.

Time Complexity:

Best case:

The target is first element,

Comparison = 1

Time complexity = $O(1)$

Average case:

The target is somewhere on mid

Avg comparison = $n/2$

Time complexity = $O(n)$

Worst case:

The target is the last element or not present

Comparisons = n

Time complexity = $O(n)$